

Chapter 13

Fostering the ‘Promising Student’ at the Outset. The Digitization and Management of Student Success in the Competitive University



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13.1 Introduction

The recent publication “The Future of Universities Thoughtbook” (FUT, 2018), a collection of forty short texts, supported by the European Commission, deals with the question “how engaged and entrepreneurial universities will drive growth and shape our knowledge-driven future until 2040” (ibid., front cover). The book’s vision of the “university 4.0” is that “academics and students work in real time symbiotic partnerships with industry, government and societal stakeholders to simultaneously create and implement new knowledge and solutions to address business and social issues” (ibid., 6). The future university is presented as a nodal point in a well-aligned network of knowledge-driven development that connects the university with industry, government, and societal agents. Within this narrative of knowledge-driven developments, “academics and students” are depicted as the university’s driving force. While the aim of universities regarding their students has been to provide them with higher education to develop their professionalism, the vision for the “university 4.0” sees them as nodes in a dynamic network of knowledge production (see also Minerva, 2020). “Students and academics” should become drivers of innovation for the society of knowledge. The envisioned university is depicted as a sphere that processes innovation through manifold connections and relations. Higher education learning is presented as “flexible, collaborative, project or

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challenge-based and cross-disciplinary, allowing students an active role in the design of their educational experience” (ibid., 11).

Visions of this kind can be found in many policy texts published in recent years, particularly those driven by the think tanks of international organizations.¹ While they claim to describe an anticipated future development, they rather sketch a ‘big picture’ of metaphors and desires for a set of reforms and policies that have transformed the university in the last decade. In these visions the university plays a crucial role in Europe becoming “the most competitive and dynamic knowledge-based economy in the world” (European Union, 2000). The concepts of entrepreneurship and competitiveness are supposed to shape the desirable behavior of individuals and organizational entities in order to activate and increase their performance. Recently, these concepts have been linked to the ideas of interconnection, datafication and digitization. It is noticeable that the vision of a “university 4.0” – as mentioned in the Thoughtbook – is put in concrete terms with respect to the students. Their essential role in the framework of competition is reflected in the question of how universities can attract the ‘best students’. This competitive task differs from recruiting the best academic staff because the choice of students has consequences for their performance. Universities have to find the most ‘promising student’.

This chapter will investigate the ways that students are addressed and constituted as innovative subjects who are supposed to take part in the competitive university. Our particular focus will be online assessment tools that are used by universities to provide students with information on study programs and advice on their choice of study options. These *online self-assessment tools* or *online study choice checks* (in German abbreviated as ‘OSA’ – we will use this acronym here) are somewhat idiosyncratic tools established in the governance of German universities in the past two decades. They are a governmental strategy to engender and generate the idea of a ‘promising student’. In this chapter, we will analyze them as an important instrument of subjectification in the geopolitical arena of knowledge-based economies. OSAs are used by universities both to develop and present a certain profile to prospective students and to recruit and select the ‘best suitable’ students to university programs. Therefore, OSAs are processors of opportunity and guiding instruments. They provide a platform of self-presentation and marketing for faculties and departments. As a consequence, they perform the act of matching students and the university as an organization and they call on them both for transformation and optimization.

We will go into greater detail on the structure of these tools at a later point in the chapter, but here we offer an initial introduction. OSAs are mostly implemented as digital tools that contain questionnaires relating to areas of interest, previous experience and, most notably, expectations that the prospective student might have. They also include tests of cognitive skills. At some universities, the completion of an OSA is optional; in other institutions it is a prerequisite for being admitted to a program. We will predominantly, though not exclusively, refer to the usage of OSAs

¹ See also the chapter by Parreira do Amaral, Chap. 3, in this volume.

at German universities, where they have come into wider use during the past ten years.² It is important to mention that, in Germany, admission to a university is not dependent on a general aptitude test as in some countries. Enrolment depends only on success in secondary education. OSAs are introduced as instruments of regulation in the transition from secondary to university education. We will demonstrate in what follows that the way they work is based on self-reflection and self-evaluation, and thus they are crucial for establishing competitiveness in the university.

Our aim in this chapter is to reconstruct OSAs as an essential element within a competitive geopolitical framing of higher education. We will investigate how the digital formation of the OSAs connects the choice of study programs to algorithmic rationalities and how their attempt to target promising students has to be seen in context of the geopolitical regime of competition in higher education. In this regime, students – as well as student satisfaction, their academic achievements, and their compliance with the program – are an indispensable resource and medium of competition.

In the first part of the chapter, we will elaborate how competitiveness currently forms a geopolitical apparatus that reorganizes higher education. We will show how competition and competitiveness are linked to practices of comparison, metrification as well as evidence-orientation. Our line of argument is based on Foucault’s and Deleuze’s accounts of governmentality and transformations of disciplinary power (Deleuze, 1992; Foucault, 2004). In the second part, we will turn toward the OSAs and to how they are utilized to enact the ideas of excellence, risk management as well as selection. By implementing OSAs, universities and prospective students ‘invest themselves’ in order to enable innovative futures. On this basis, we will delineate the subject-formation of being a ‘successful’ or ‘promising’ student in the third part of the chapter. We will argue how – according to the measures of digitization and learning analytics – the differential of digital self and digital optimum addresses prospective students to become an ‘entrepreneurial academic subject’. The chapter concludes with a reflection on the educational consequences of these transformations for higher education.

13.2 Competitiveness in the University and in Higher Education

In his recent contributions on the geopolitics of the knowledge-based society, Moisio (2018) emphasized the connectivity that “becomes a crucial resource in the emerging ‘global network civilization’ in which ‘mega-cities compete over

²Hell et al. have counted more than six hundred self-assessment tools in Germany (Hell et al., 2018, 133). There is a wide range of types and forms of OSAs. In recent years, there have been attempts to collect them on a single platform (see www.osa-portal.de; cf. Höft and Hell 2017). The wide range of tools is as confusing and diverse as the study programs: In 2019, there existed roughly 20,000 study programs in Germany (see HRK, 2019).

connectivity’ and in which state borders are increasingly irrelevant” (ibid., 5; see also Moisis, Chap. 2, in this volume). Connectivity becomes relevant for universities, which are increasingly involved in the construction of geopolitical arenas. The connecting practices might be instituted regionally or nationally – as for instance in the excellence programs for universities in Germany – but they are also constituted on a global level. To set the connected entities in competitive relations, rankings and other metrical practices is essential to this. Evaluations, the emergence of a new and all-encompassing audit culture (see Spooner 2017), etc. lead universities to understand themselves as neoliberal competitors who rival others through the enactment of their own efforts at optimization; a rivalry that is quantifiable, measurable and correspondingly comparable (Dimitrova & Dimitrova, 2017). Erkkilä and Piironen (2020) demonstrated recently how university rankings foster competitiveness on a global scale: “[T]he global university rankings help to transfer the global imaginaries of competition to regional and local level” (ibid., 39).³ Rankings provide the data and assessment criteria to make policy intervention possible.

The use of rankings brings about a changed normative framework by transforming the respective organizations into competitors: Institutional routines and processes as well as mission statements have to be changed accordingly. For the area of research, Bröckling and Peter (2017) have demonstrated that ‘excellence in the university’ is linked to the task of presenting oneself as excellent and of permanently working on one’s own self-optimization. One of the central elements of competition is the technology to “discover” excellence by constructing a market environment that operates on the grounds of indicators and rankings (ibid., 292–294). The excellence program in Germany or the Research Excellence Framework (formerly Research Assessment Exercise; RAE) in the UK are examples of technologies that constitute these market environments (for the RAE see Schäfer, 2019): Researchers in a particular field of research have to compete with their colleagues in terms of their publications and even their future research plans: Who has the most innovative ideas? As the academic organizations in the connected network are their own review bodies, they form a tribunal in which they are both the judges and the judged. This recursive relation increases the identification with the self-established criteria of excellence.

In the realm of teaching in higher education, there have also been developments toward competitiveness, but here the formation of an environment of competition differs: The competitive advantage of a university is rooted in the possibility that the university can offer its students a successful course of study. There is a paradoxical relation in this possibility: Universities compete over the student’s potential future effort. For some time, universities have collected data on student dropouts and the time that students need to successfully complete their study program. With the advance of datafication and the rise of so-called ‘learning analytics’, it becomes possible to analyze ‘correlations of success’, i.e. to make out the student features that will most likely lead to a successful course of study. With these digitized

³ See also Boyadijeva, in this volume.

practices, the universities gain capability to manage the uncertainty of student success. One way to do so is to use the collected data for the elimination of contingency and uncertainty (see Amos, 2019).

The identification of the 'promising student' is not simply a matter of finding the naturally given talents, however. Rather, it means to address students with respect to their potential success through self-evaluation and self-optimization. It is a particular way of 'being a student' – the desired way in a competitive university – and this is where the study choice tools come into play: They are rather instruments of self-exploration and the point of departure for a promising future.

The governmental subject-formation of the 'promising student' has to be seen in close connection to today's demands for permanent development, production, and self-improvement (Foucault, 1993, 2004; Deleuze, 1992). Disciplinary forms of social control are increasingly replaced by a "progressive and dispersed installation of a new system of domination" (Deleuze, 1992, 7), which installs "ultrarapid forms of free-floating control" (ibid., 4). These "data-based" (Bächle, 2016, 164) "societies of control" operate through continuous variations and numerical modulations (Deleuze, 1992, 4). Deleuze also points out that educational institutions such as the university are handed over "to the corporation" (ibid., 5). Clearly, this implies an enormous potential for control: Whereas – in earlier times – individuals were time and again released from a particular institutional formation (e.g., after an examination), within societies of control there is no dismissal anymore (ibid.). Following Deleuze, this control is also related to the connectivity between different areas and institutions and we read this as a particular manifestation of the geopolitical transformation of higher education. Take e.g. the ways that universities and corporations are linked by cooperative study programs or by 'service learning'.

According to Deleuze, institutions are transformed from sturdy patterns to "continuous network[s]" (ibid., 6). The orientation towards corporations requires flexible compounds and sustained ways of networking and consolidation that makes any notion of stagnation or final destination impossible. As Deleuze puts it: "The family, the school, the army, the factory are no longer the distinct analogical spaces that converge towards an owner – state or private power – but coded figures – deformable and transformable – of a single corporation that now has only stockholders" (ibid.). Even though we are skeptical of Deleuze's totalizing concept of "corporation", we think that his references to deformation and modulation are helpful to understand current changes in the domain of higher education where study programs, university degrees, as well as student experiences become accessible for modulation and governmental subjectification (in this context see Ott, 2015).

It does not come as a surprise that quantification and new forms of conduct through numbers are particularly relevant to constitute these networks. They are the reference point by which stakeholders relate to one another. Wæraas and Solbakk (2009) even point out – following Bleiklie and Kogan (2007) – that the modern university can be characterized as a "stakeholder university" rather than an academic community (Wæraas & Solbakk, 2009, 450). The connectivity between different stakeholders – e.g. students, teachers, but also "governmental agencies" (Greller & Drachsler, 2012, 46) or "commercial service providers and even

automated [computer] agents” (ibid., 44) – transforms the university into a geopolitical nodal point of opportunities. However, one also has to acknowledge that the production of data and numbers becomes more and more relevant to drive and transform the competitive efforts of the university. Put differently, quantification becomes a medium and driving force for governmental rationalities. In order to delineate more closely how the governmental subject-formation or subjectification of ‘student experience’ or ‘academic subjectivity’ operates in line with this development of quantification, we will turn to the OSAs as one very important instrument that universities employ for managing student success.

13.3 OSAs: On the Management of Risk and Student Success

The topic of admission to universities is a complex issue that equally touches upon economic, political, and cultural conditions of academia and is a central concern for university governance. Depending on these conditions, the admission to a program can be dependent on general aptitude tests, on recommendation letters (mostly in M.A. programs), on application essays, etc. Some instruments design or construct the choice of study as a self-reflective process. This is the case in ‘admission interviews’, for example, where prospective students are asked to reflect on their motivations for study. The OSAs are another example of addressing prospective students with respect to their compatibility with the study program. The example of Saxion University in the Netherlands illustrates how the study choice counseling operates.

In a 2-min Youtube video, the university informs international students about the admission procedure (Saxion, 2020). The friendly voice in the videoclip first informs the viewer that the “study choice check” is not an entrance exam and does not have any influence on the admission process. The voice continues: “At Saxion, we believe that making the right choice when it comes to your study choice is very important” (ibid.). The “study choice check” is seen as the very instrument that allows prospective students to make this “right choice”: “With this check you get a good idea whether the program will be a good fit for you” (ibid.). The instrument makes use of assignments appropriate to the study program (e.g., case study) and – depending on the program – combines this with an advice session. The final image shows a wooden toy known as a “shape sorter”. Three possible results are distinguished in the video clip: a full match between student and program; a sufficient match (i.e., with advice to put in some extra effort in some areas); and no match, followed by the explicit advice that it would be better to “choose a different degree program” (ibid.).

As can be concluded from this short example, the ‘study choice check’ has to be seen as an advice instrument that redefines the transition to the university from a collective appointment (remember the inauguration lectures given by scientific personalities such as Horkheimer to first-year students) to an individualized situation

of choice and self-reflection. Ott et al. have described OSAs as “a form of diagnostics that one applies to oneself” (Ott et al., 2017, 217, our transl.). When analysing these and similar programs, progress reports from universities, and policy documents from the German context, the strategic significance of the OSAs becomes evident. In what follows we will emphasize three aspects: the discovery of talent; the reduction of costs and risks; as well as the refinement of the university profile.⁴

As mentioned before, OSAs are developed by universities to foster the success of students by improving their decision-making about what to study (Heinitz, 2017). In 2004 the German advisory council of research and science (*Wissenschaftsrat*) recommended the implementation of “aptitude assessment procedures for the purpose of counselling” in order to improve the match between students’ entry qualifications and requirements of the study program in advance of admission (see *Wissenschaftsrat*, 2004, 6). The idea was to set an expectation that prospective students would reflect on their suitability for a study program and thus reduce universities’ dropout rates (*ibid.*, 4).

It is mentioned time and again that the OSAs are the best way to find suitable applicants for a university’s study program (Ott et al., 2017, 221). Even though the results may not play a role and there is no obligation to follow the results or advice of the OSAs, the proponents of this instrument explicitly and deliberately count on processes of self-selection, i.e., that students with negative results will change their plans or alternatively compensate their deficits. More particularly, the aim is to come to a “well-founded self-selection” (Thiele & Kauffeld, 2019a, 275, our transl.). However, the OSAs are also geared towards finding the best-fitting students: The OSAs are required to be “resource-oriented” and not merely “deficiency-oriented” (Wolff-Grosser, 2018, 93, our transl.).

This last remark – as with the above-mentioned Youtube example from Saxion University – shows that the OSAs cannot be reduced to a method of ‘limitation of access’ to the university. Rather, the idea of the ‘match’ and the promised harmony between student and program redefines the ‘location of study’. In the documents we find the notion of an “agreement in values” (Thiele & Kauffeld, 2019a, 275; cit. Heise et al., 1999), the “perfect fit” between person and study environment, which will lead to, among other things, a higher student satisfaction and study success rate (Thiele & Kauffeld, 2019a, 275). The university offers, in other words, the perfect environment to bring students on their way and realise their potential.⁵ Accordingly,

⁴In this chapter we cannot provide a full discourse analysis of the collected text material. Our aim is to identify central motifs or aspects of the strategic relevance given to OSAs. The text material mostly stems from progress reports from universities as well as programmatic texts from university policy. Furthermore, it also contains how-to-instructions or manuals for the implementation of an OSA at a university (e.g. Weber et al., 2019) as well as critical self-reflections concerning the scientific standards of expertise and evaluation. We will not discuss the OSAs with respect to their quality or evidence. Existing empirical studies (see for instance Dietrich et al., 2019; Höft et al., 2019; Röder, 2017) show that many universities provide insufficient OSAs: Only 61% of the analyzed tests were evaluated adequately (Thiele and Kauffeld 2019b, 127).

⁵Simultaneously, the university can reduce the risk of investing resources without return (see Schröder et al., 2018).

the ‘location of study’ is personalized: With the help of OSAs, the universities do not remain “anonymous educational institutions” (ibid., 234). The OSAs can establish loyalties and are an active sight of “educational branding” (Adler et al., 2019, 41).

13.4 Student Subject Formation and the Quest of Data

In the previous part of the chapter we demonstrated how OSAs are implemented in order to produce and manage the uncertainty of study success. By providing prospective students with information on how far their interests and preceding experience match the study program, they learn whether their expectations correspond to the respective study program. By performing the OSAs, prospective students become aligned with the vision of successful study. The universities provide materialized hints to show exactly what they aspire to: the idea of an ideal-student that fits the program perfectly. At the same time, prospective students are asked to examine themselves according to this ideal.

The quantification of ‘auspicious starting conditions’ is made possible by collecting and analyzing data. This opens the possibility for a ‘responsible study choice’. The more data that is available, the more explanatory power will be attributed to the insight into the ‘black box’ of the prospective student. With respect to educational data mining (EDM), Behrens and DiCerbo speak of a shift from the “digital desert” to a “digital ocean” where the abundance of digital data will revolutionize the nature of human intellectual and cultural history (Behrens & DiCerbo, 2014, 39f.). According to Bächle, it is this abundance of data that advances the power of the societies of control by simultaneously promising a total “legibility of the self through data” (Bächle, 2016, 172; our transl.).

In the present context, the problem definition that underpins OSAs within higher education is that many students do not fit the chosen program and drop out. Hence, there has been extensive research to identify the indicators of student failure – Antunes (2010) speaks about “ASAP classifiers” that could be implemented to automatically predict a student’s probability of success or failure in order to implement countermeasures at the earliest stage possible. The OSAs are a very interesting alternative in this regard: “Problem students” can be identified even before entering the program. Furthermore, the OSAs foster a ‘data perspective’ on students because they address a wider range of student characteristics. These support the construction of “learner models” in order to make out the predictors of study success (cf. Pea, 2014).

Recent developments in learning analytics and educational data mining have nourished the idea that student learning can be technologically enhanced (Larsson & White, 2014). The idea is that data makes special needs and possibilities of optimization accessible so that educational interventions can be implemented with respect to the individual student (Greller & Drachsler, 2012, 54). Amos (2019) gives an example from the University of Arizona, which has implemented an e-advice

system to increase retention rates. A counselor is notified when the algorithmic automated tracking notes a range of 'problematic student behavior', such as not completing coursework on time or spending more hours in the gym than in the library (ibid., 238). It is here that we can see how preemptive and predictive strategies of governance merge in student subject formation.

Data correlations on student performance restructure the task of higher education quality management; for it changes what it means to study. Companies such as Abstract Technology, for example, establish architectures to analyze learning experiences (see the open edX platform: AT 2020). These digital architectures aggregate student data from multiple courses and use the resulting analyses to rearrange course material according to the weaknesses of students. In other words, the data is used to optimize the learning environment. As Mau (2019) has argued with respect to big data, the quantification process goes along with a valorization (see also Heintz, 2010). Instead of merely representing the world (a certain student competence), it creates a new reality with hierarchical inequalities (Mau, 2019, 40–59). In the case of student course work this might be the idea of 'committed students' who put in the time recommended for the course they have signed up for. More specifically, the value manifestation is associated with the student's attitude, with an 'active disposition' toward ideal-student-subject-formation. Higher education's adaptation of strategies of 'algorithmic governance' means OSAs might form the entrance point for an entirely new 'digitized study culture'.

One important aspect of student subject formation is that the data constructs a "digital self" or a "statistical artefact" (Bächle, 2016, 194; our transl.) by collecting and evaluating information on their performance (e.g. test results, or general user behavior on learning platforms). Instead of viewing this as 'insight into a black box' – as it is frequently referred to – it is important to acknowledge the constructive dimension of the data with respect to the "digital self". Critical accounts of quantification and of the misconceived idea that data can "speak for themselves" (Kitchin, 2014, 5) or that measurement "provides privileged or exclusive access to the real" (Espeland & Stevens, 2008, 432) by delivering objective knowledge should be borne in mind here. From a posthuman perspective, Goriunova (2019) has pointed out that the digital subject is constructed through matching, correlating and modeling and that digital subjects are produced 'at a distance'. We use the term "digital self" here in order to indicate that the data *are associated* to a particular self by way of subjectification. In other words: The collected data provide an attachment by suggesting that the numbers and indicators can be attributed to the respective individual: "My OSA result was ..."

Put this way, data do not speak for themselves but rather for the data-delivering subject (Hörtnagl, 2019, 146) as an expression of a measured 'algorithmic identity'. The example of an identity document can be used as an illustration of this translation: the subject becomes visible through its constitutive data (Bächle, 2016, 188). To speak of a constitution also implies a translation – it could be described as an insufficiently complex fragmentation of the subject rendered by processing only the elements needed for the digital modelling of a user's interface or to create profiles (Hartong, 2019, 12).

We can see here Deleuze's dictum, that subjectification is accomplished through modulation: By using learning platforms, OSAs etc. it is possible to collect and automatically analyze data about students' interests, expectations, motivational and cognitive skills as well as to assemble them into a specific 'picture'. Besides the construction of a 'digital self', it also connects modulations of future and risk and, as a consequence, supports the dataveillance constitutive of societies of control: By producing categories and classification of specific pictures – e.g., a label of the soon-to-be-failing-student – it will be possible to tag them with presumption and risk (see also Bächle, 2016, 188) and to counteract their 'failure' preemptively.

What is also relevant when investigating the effects of OSAs is that the data generates a *digital optimum*. As algorithmic rationality is continuously applied to the learning or testing environment, it simultaneously optimizes and normalizes the processes and results of evaluation. With respect to the OSAs, this means that prospective students are confronted with the ideal of a 'perfect start' for their studies. The OSA transmits specific expectations in terms of programs or intramural performances and students are faced with the demand to make this compatible with their own expectations. Students may, for example, find out in the OSA that the lack of a language stay abroad means a poor starting position for studying a foreign language. They will take it as *their* responsibility to compensate for this disadvantage. On the basis of their OSA results, students will start to manage their study portfolio. There is an interpellation of data that calls for risk management and self-optimization.

In sum, digital self and digital optimum form a differential that exerts algorithmic governance (Beer, 2016). This differential shapes student behavior (also in relation to other students) and thus it restructures higher education. In our conclusion we would like to highlight the changes that these developments can bring about for the culture of studying and of higher education in general.

13.5 The Transformation of Higher Education and the Exercise of Critique

In their book "Global Immunity", Jan Masschelein and Maarten Simons (2005) delineated the changes brought about by the creation of the European Space of Higher Education, and pointed toward the role of a "quality tribunal" that makes everyone accountable in the university (see also Masschelein & Simons, 2010b). In this chapter, we have focused on contemporary developments in higher education that fit very much this diagnosis. In the first part of the chapter we demonstrated how competition and competitiveness within higher education is closely linked to the possibility to 'manage' student success. We have also shown that this field of governance is currently being restructured in light of the possibilities afforded by learning analytics and big data. In this context, the instruments of 'study choice check' or OSAs have a prominent role: They are an instrument with which to

identify preemptively problems within the course of study for prospective students. Furthermore, they change the idea of what it means to be a 'good' or 'promising' student.

The techno-social imaginary of a perfect learning environment is connected to a strong individualization of higher education. In this regard, our analysis confirms what Masschelein and Simons have labeled a "trajectory of learning" for every individual, where everyone is on their own path of learning (Masschelein & Simons, 2010a, 669). We have placed a strong emphasis on the subjectifying effects of performance data collection, i.e., the call for optimization addressed towards students in the form of permanent progress assessment, which also concerns the study programs and the responsible academic departments. The competitiveness and the geopolitical advantage of a university will depend strongly on the possibility to manage student success appropriately.

This vision of the perfect governance of learning comes with constraints and reductions, however. We follow Kitchin in his diagnosis that subjects and data are "always in a state of becoming"; they are "uncertain, provisional and messy fragile" (Kitchin, 2017, 18). The current approaches lack the fundamental idea within (higher) education that education and *Bildung* are related to an open and unpredictable future; they go beyond the correlations of data (see Jornitz and Klinge, Chap. 14, in this volume). The logic of representation in the relation of subject and data, and the idea of predictability, undermines this potentiality. The students' future is reduced to their 'predicted future'. It is this vision of study success based on personalizing the learning experience that we have to interpret as "algorithmic governmentality's boundless strength" (Berns, 2018, 256).

Universities are places where academic communities construct knowledge and perform deliberation, and these practices are dependent on a common space in which the members of the university can engage with each other. What happens if everyone simply moves in one's own learning environment? We suspect that the openness of academic exchange will diminish when study success becomes a matter of prediction and aggregated probabilities, for there will be nothing really to expect from engagement with others. When moving along a learning trajectory one is concerned only with one's own path of optimization.

In our view this will limit tremendously the scope of higher education, which has to include those more troublesome aspects of study experience that digital technology seeks to mitigate against. The delays in reading and writing, e.g., can be seen as a barrier as well as a *resource*: when a question is not yet ready to be asked, this might not be a failure of achievement. Higher education also requires a critical engagement with the academic discipline: Students become professionals in and through taking a stance toward the knowledge they are confronted with. In other words, higher education is about partaking in the reflective evaluation of what is presented *as* knowledge. It is critique that is at stake here and that has been valued in critical traditions from Kant to the Frankfurt School to Foucault. The exercise of critique is a crucial knowledge practice within the course of study that reopens what is there to think and to explore.

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